



ROBERT BOSCH
(AUSTRALIA) PTY. LTD.
A.B.N. 48 004 315 628

ENGINEERING SPECIFICATION

FOA Barra Body Electronics Module
High and Low Series

F005 VP 0058

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10.1 Module Aliases: Barra BEM

FORD AUTOMOTIVE OPERATIONS SUBSYSTEM SPECIFIC DIAGNOSTIC SPECIFICATION (Part II) for the BARRA BODY ELECTRONICS MODULE (BBEM)									
Module Aliases: Barra BEM				PART II - RELEASE DATA					
ECU (Module) DATA				PART II - RELEASE DATA					
MRDB Acronym	Node Hex Address	Diagnostic Com. Link	Supplier	Part II Suffix	WERS Notice #	Design & Release Engineer			
						Name	PROFS ID	Phone	
GEM	Rx \$726 Tx \$72E	HS-CAN (500K)	Robert Bosch (Aust.) PL			Manickam Sivasothy (Siva)	RBAU-EGE3-MS	+61-3-9541 5160	
						Matthew Brooks	RBAU-STF-MB	+61-3-9541 5343	
Gateway Node Data									
Gateway (Y/N)	Host Address	Host Acronym	Host Com. Link						
NO									
Diagnostic Requirements				Part Number					
Global Diagnostic Specification (Part 1)				Version 2001.0 16-Feb-01					
Module Programming / Configuration Design Spec.				Version 2001.0 9-Mar-01					
CAN Generic Diagnostic Specification				Version 2001.0 9-Feb-01					
CONTACTS									
Web Release Area (URL): http://eesftp.pd6.ford.com/diag/webpage									
Diagnostic Application Engineer (Ford EESE):									
				Ross Bunting	FOA	rbunting@ford.com	+61-3-9359 7573		
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				Bharat Vyas (B.V.)	FOA	bvyas1@ford.com	+61-3-9359 7078		
APPLICATION DATA (Usage)									
Program Usage		Part Number Stored in ECU (PIDs)			Type		ECU Description		WERS Notice #
Code(s)	Model Year(s)	Prefix (\$E21A)	Base (\$E217)	Suffix (\$E219)					
BBEM/L OE	2003	3R23	14K144	AA	F005V00207	Barra BEM Low Series Original Equipment			
BBEM/H OE	2003	3R23	14K144	BA	F005V00222	Barra BEM High Series Original Equipment			
BBEM/L PA	2003	3R23	14K144	AA	TBA	Barra BEM Low Series Parts & Accessories			
BBEM/H PA	2003	3R23	14K144	BA	TBA	Barra BEM High Series Parts & Accessories			

10.2 Messages & Services Supported

MESSAGES & FUNCTIONS SUPPORTED				for additional information, reference: Appendix A - Message Structures & Responses of the Global Diagnostic Specification (Part 1) - Worldwide Requirements			
Form version 4.0							
Supported							
Mode	Message Description	Y/N	Comments / Information	ECU Type			
10	Start Diagnostic Session	N	Not supported.	-	-	-	-
11	ECU Reset	Y		-	-	-	-
12	Read Freeze Frame Data	N	Not supported.	A	B	-	-
14	ClearDiagnosticInformation	Y		-	-	-	-
18	ReadDTCByStatus	Y		A	B	-	-
21	ReadDataByLocalIdentifier	Y		A	B	-	-
22	ReadDataByCommonID	Y		A	B	-	-
23	ReadMemoryByAddress	N	Not supported.	-	-	-	-
24	Req Common ID Scaling Masking	N	Not supported.	-	-	-	-
27	Security Access	N	Not supported.	-	-	-	-
28	CommunicationControl	Y		-	-	-	-
2E	WriteDataByCommonID	Y		A	B	-	-
2F	Input Output Control By Common ID	Y		A	B	-	-
31	Start Routine By Local Identifier	Y		A	B	-	-
32	Slop Routine By Local Identifier	Y		A	B	-	-
33	Request Routine Results By Local ID	Y		A	B	-	-
34	Request Download	N	Not supported.	-	-	-	-
35	Request Upload	N	Not supported.	-	-	-	-
36	Transfer Data	N	Not supported.	-	-	-	-
37	Request Transfer Exit	N	Not supported.	-	-	-	-
3B	Write Data By Local ID	Y		-	-	-	-
3D	Write Memory By Address	N	Not supported.	A	B	-	-
3E	Tester Present	Y		-	-	-	-
85	Control DTC Setting	Y		A	B	-	-
A0	Request Diagnostic Data Packet	N	Not supported.	A	B	-	-
A1	Diagnostically Define Data Packet	N	Not supported.	-	-	-	-
B1	Diagnostic Command	N	Not supported.	-	-	-	-
Supported							
Number	Function	Y/N	Comments / Information	ECU Type			
1	Gateway Mode	N	Not supported.	-	-	-	-

Issue Number:

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Valid from:

23 September 2002

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10.3 Diagnostic Routines – On Demand Self Test

DIAGNOSTIC ROUTINES		for additional information, reference:			
Form Version 4.0					
TEST NAME:	On-Demand Self-Test				
TEST NUM.:	\$02	EXECUTION (Actual) TIME:	14 seconds		
TEST DESCRIPTION					
Pre-condition: Key in ignition position, headlamp switch in "auto" position, and boot is shut, glovebox open.					
Visual Observation: Battery Saver (glovebox lamp), Heated Back Light, passenger door unlock and washer motor tests require visual observation. If no water in washer bottle then listen for motor operation.					
Manual Operation and Visual Observation: Power Window test requires holding the button and observing window operation.					
ECU TEST ENTRY Requirements			Comments / Information		
Receipt of "REQUEST DIAGNOSTIC ROUTINE ENTRY" message containing the above test number.					
Use startRoutineByLocalIdentifier (Service ID \$31)					
ECU TEST EXIT Requirements			Comments / Information		
ECU does not receive a Node-to-Node message every 5 seconds.					
REQUEST RETURN TO OPERATIONAL STATE message received					
Receipt of "REQUEST DIAGNOSTIC ROUTINE EXIT" message (unconditional).			Exit immediately		
Conn. Pin Num.	I/O	Circuit Description	Circuit Evaluated During Test (Y/N) Setup / Entry Requirements	Comments / Information	ECU Type
J2-24, J5-2	O	Driver Door Lock	Y -	Relay energised for 200ms. (locks all doors). Feedback controls DTCs, DTC \$9300	A B
J2-24, J3-9	O	Driver Door Unlock	Y -	Relay energised for 200ms. Feedback controls DTCs. DTC \$9982	A B
J2-24, J5-1	O	All Door Unlock	Y -	Relay energised for 200ms. Visual inspection. DTC \$9982	A B
J2-24	I	Driver Door Lock feedback	Y -	Feedback is used to raise DTCs \$9300 and \$9982	A B
J2-22	O	Seat Module Data Line - Open circuit/SC to ground	Y -	DTC \$A073	- B
J2-22	O	Seat Module Data Line - Short to Battery	Y -	DTC \$A072	- B
J2-23	O	Seat Module Enable Line - Open circuit/SC to ground	Y -	DTC \$A075	- B
J2-23	O	Seat Module Enable Line - Short to Battery	Y -	DTC \$A074	- B
J2-9	O	Horn Relay Line - Open Circuit	Y -	DTC \$A849	A B
J2-9	O	Horn Relay Line - Short to battery	Y -	DTC \$A511	A B
J3-10	O	Front wash motor	Y Key in Ignition position	DTC \$9458	- B
J3-3	O	Battery Saver	Y -	Leave open "glove box" or turn on "map lamp" to observe.	A B
J1-8	O	Transceiver TX Data Line - Open circuit	Y Battery Saver Active	DTC \$A930	A B
J1-8	O	Transceiver TX Data Line - Short to Battery	Y Battery Saver Active	DTC \$A929	A B
TILA	I	TILA Challenge - Key ID unknown	Y Battery Saver Active, Key in Ignition Position	DTC \$9301	A B
J4-4	I	TILA Challenge - Faulty transponder	Y Battery Saver Active, Key in Ignition Position	DTC \$9982	A B
TILA	I	TILA disconnected	Y Battery Saver Active, Key in Ignition Position	DTC \$A103	A B
TILA	I	TILA Challenge - Faulty or Fixed code transponder	Y Battery Saver Active, Key in Ignition Position	DTC \$9302	A B
CAN Msg \$6F6	I	EEC (PCM) Communication Fault	Y Battery Saver Active, Key in Ignition Position	DTC \$A682. CAN message \$6F6 missing	A B
J3-1, J3-2	I/O	Front wiper motor - Wiper feedback	Y Key in Ignition position	1 wipe, DTC \$9446	A B
J2-19	I	Front wiper switch potentiometer circuit failure	Y Key in Ignition position	DTC \$A891	A B
J3-7, J1-17	I/O	Boot Release / feedback	Y Battery Saver Active, Boot needs to be shut	Relay energised for 800ms. DTC \$A894	A B
J4-8, J4-9	I	Cabin Temperature Sensor - Circuit Open	Y Key in Ignition position	DTC \$9251	- B
J4-8, J4-9	I	Cabin Temperature Sensor - Short to Gnd	Y Key in Ignition position	DTC \$9253	- B
J4-8, J4-9	I	Cabin Temperature Sensor - Out of range	Y Key in Ignition position	DTC \$9250	- B
J2-6, J2-17	I	Driver Sunload Sensor - Circuit open / Short to battery	Y Key in Ignition position	DTC \$9259	- B
J2-6, J2-17	I	Driver Sunload Sensor - Short to ground	Y Key in Ignition position	DTC \$9251	- B
J2-10	O	Heated Backlight Relay	Y Key in Ignition position	Look at dash board to observe indicator	A B
J2-10	O	Heated Backlight Relay - Open Circuit	Y Key in Ignition position	DTC \$9348	A B
J2-10	O	Heated Backlight Relay - Short to battery	Y Key in Ignition position	DTC \$9349	A B
J2-11	O	Power Windows	Y -	Hold button to observe window operation	A B
J2-11	O	Power Windows - Open Circuit	Y -	DTC \$9548	A B
J2-11	O	Power Windows - Short to Battery	Y -	DTC \$9549	A B
J3-12	O	Dome Lamp	Y -	Visual Inspection	A B
J1-18	I	Twilight Sensor - Circuit Open / Short to battery	Y Battery Saver Active	DTC \$9791	- B
J1-18	I	Twilight Sensor - Out of range	Y Battery Saver Active	DTC \$9790	- B
J1-5,6	I/O	Auto Headlamp Feedback	Y Headlamp switch in "auto" position	DTC \$9893	- B
EEPROM	-	Number of transponder keys programmed is less than minimum	Y -	DTC \$9213. Less than 2 transponders programmed.	A B
CAN Msg \$207	I	CAN bytes for vehicle speed is read as "FF FF"	Y Key in Ignition position	DTC \$E197	A B
CAN Msg \$427	I	Airbag Deployed CAN bit is received prior to or during self-test.	Y Key in Ignition position	DTC \$9543	A B
TILA, J1-7	I	TILA RF receiver error	Y -	DTC \$9900	A B
SYSTEM	-	Key programming failed	Y -	DTC \$A431. Key programming not checked during self-test	A B
CAN Bus	I/O	CAN Bus off error detected	Y Key in Ignition position	DTC \$C903. Used with PID \$6902	A B
CAN Msg \$6F6 EEPROM	I	Cabin Sensor calibration factor is not configured.	Y -	DTC \$A477	- B
CAN Msg \$6F6 EEPROM	I	No PCM ID Stored	Y -	DTC \$A141	A B
INTERNAL	-	EEPROM / ROM / RAM / TEMC / Autoheadlamp Failure	Y Key in Ignition position	DTC \$9342	A B

Note:

CAN connector pins are J2-13, 14, 15, 16.

TILA connector pins are J1-3, 4, J1-7, 8.



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10.4 Diagnostic Routines – Assembly Self-test

DIAGNOSTIC ROUTINES		for additional information, reference:			
Form Version 4.0					
TEST NAME: Assembly Self-Test					
TEST NUM: \$11		EXECUTION (Actual) TIME: 6.5 seconds			
TEST DESCRIPTION					
Pre-condition: Key in Ignition position, headlamp switch in "auto" position, and boot is shut.					
Visual Observation: Omitted in Assembly Self-Test					
ECU TEST ENTRY Requirements					
Receipt of "REQUEST DIAGNOSTIC ROUTINE ENTRY" message containing the above test number.		Comments / Information			
Use startRoutineByLocalIdentifier (Service ID \$31)					
ECU TEST EXIT Requirements					
ECU does not receive a Node-to-Node message every 5 seconds.		State Exited To			
REQUEST RETURN TO OPERATIONAL STATE message received		Default / Diagnostic State \$81			
Receipt of "REQUEST DIAGNOSTIC ROUTINE EXIT" message (unconditional).		Default / Diagnostic State \$81			
		Exit Immediately			
Conn. Pin Num	I/O	Circuit Description	Circuit Evaluated During Test (Y/N) Setup / Entry Requirements	Comments / Information	ECU Type
J2-24, J5-2	O	Driver Door Lock	Y	Relay energised for 200ms (locks all doors). Feedback controls DTCs, DTC \$9300	A B
J2-24, J3-8	O	Driver Door Unlock	Y	Relay energised for 200ms, Feedback controls DTCs, DTC \$9302	A B
J2-24, J5-1	O	All Door Unlock	Y	Relay energised for 200ms. Visual inspection, DTC \$9302	A B
J2-24	I	Driver Door Lock feedback	Y	Feedback is used to raise DTCs \$9300 and \$9302	A B
J2-22	O	Seat Module Data Line - Open circuit/SC to ground	Y	DTC \$A073	- B
J2-22	O	Seat Module Data Line - Short to Battery	Y	DTC \$A072	- B
J2-23	O	Seat Module Enable Line - Open circuit/SC to ground	Y	DTC \$A075	- B
J2-23	O	Seat Module Enable Line - Short to Battery	Y	DTC \$A074	- B
J2-9	O	Horn Relay Line - Open Circuit	Y	DTC \$A849	A B
J2-9	O	Horn Relay Line - Short to battery	Y	DTC \$A511	A B
J3-10	O	Front wash motor	Y	DTC \$9458	- B
J3-3	Q	Battery Saver	Y	Leave open "glove box" or turn on "map lamp" to observe.	A B
J1-8	O	Transceiver TX Data Line - Open circuit	Y	DTC \$A830	A B
J1-8	O	Transceiver TX Data Line - Short to Battery	Y	DTC \$A829	A B
TILA	I	TILA Challenge - Key ID unknown	Y	DTC \$9801	A B
J1-4	I	TILA Challenge - Faulty transponder	Y	DTC \$9802	A B
TILA	I	TILA disconnected	Y	DTC \$A103	A B
TILA	I	TILA Challenge - Faulty or Fixed code transponder	Y	DTC \$9802	A B
CAN Msg \$6F6	I	EEC (PCM) Communication Fault	Y	DTC \$A682, CAN message \$6F6 missing	A B
J3-1, J3-2	I/O	Front wiper motor - Wiper feedback	Y	1 wipe, DTC \$9446	A B
J2-19	I	Front wiper switch potentiometer circuit failure	Y	DTC \$A891	A B
J3-7, J1-17	I/O	Boot Release / feedback	Y	Relay energised for 800ms, DTC \$A894	A B
J4-8, J4-9	I	Cabin Temperature Sensor - Circuit Open	Y	DTC \$9251	- B
J4-8, J4-9	I	Cabin Temperature Sensor - Short to Gnd	Y	DTC \$9253	- B
J4-8, J4-9	I	Cabin Temperature Sensor - Out of range	Y	DTC \$9250	- B
J2-6, J2-17	I	Driver Sunload Sensor - Circuit open / Short to battery	Y	DTC \$9261	- B
J2-6, J2-17	I	Driver Sunload Sensor - Short to ground	Y	DTC \$9261	- B
J2-10	O	Heated Backlight Relay	Y	Look at dash board to observe indicator	A B
J2-10	O	Heated Backlight Relay - Open Circuit	Y	DTC \$9348	A B
J2-10	O	Heated Backlight Relay - Short to battery	Y	DTC \$9349	A B
J2-11	O	Power Windows	Y	Hold button to observe window operation	A B
J2-11	O	Power Windows - Open Circuit	Y	DTC \$9546	A B
J2-11	O	Power Windows - Short to Battery	Y	DTC \$9549	A B
J3-12	O	Door Lamp	Y	Visual Inspection	A B
J1-18	I	Twilight Sensor - Circuit Open / Short to battery	Y	DTC \$9791	- B
J1-18	I	Twilight Sensor - Out of range	Y	DTC \$9750	- B
J1-5,6	I/O	Auto Headlamp Feedback	Y	DTC \$9693	- B
EEPROM	-	Number of transponder keys programmed is less than minimum.	Y	DTC \$9213, Less than 2 transponders programmed.	A B
CAN Msg \$207	I	CAN bytes for vehicle speed is read as 'FF FF'	Y	DTC \$E197	A B
CAN Msg \$427	I	Airbag Deployed CAN bit is received prior to or during self-test	Y	DTC \$5843	A B
TILA, J1-7	I	TILA RF receiver error	Y	DTC \$9600	A B
SYSTEM	-	Key-programming failed	Y	DTC \$A131, Key-programming not checked during self-test	A B
CAN Bus	I/O	CAN Bus off error detected	Y	DTC \$D900, Used with PID \$6902	A B
CAN Msg \$6F6 EEPROM	I	Cabin Sensor calibration factor is not configured.	Y	DTC \$A477	- B
CAN Msg \$6F6 EEPROM	I	No PCM ID Stored	Y	DTC \$A141	A B
INTERVAL	-	EEPROM / ROM / RAM / TEMIC / Autoheadlamp Failure	Y	DTC \$9342	A B

Note:

CAN connector pins are J2-13,14,15,16.

TILA connector pins are J1-3,4, J1-7,8.

10.5 Parameter Identifier (PID)

10.5.1 Numeric Diagnostic PID

PIDS										For: REPORTING: INPUT, OUTPUT & FUNCTIONAL STATUS			for additional information, reference: Section 8.0 - Parameter Identifier (PID) Requirements of the Global Diagnostic Specification (Part 1) - Worldwide Requirements				
Form Version 4.0										NOTE: PID information can be copied directly from the Diagnostic Database and pasted into this specification.							
NUMERIC PIDS - PIDs whose information is provided as Numeric data lying within a defined range.																	
										Writeable States							
Conn Pin Num	PID	PID Name	Units	Size	Data Type	Resolution	Offset	Minimum	Maximum	\$2F	\$2E	Comments/Information	ECU Type				
EEPROM	0200	Number of Continuous Trouble Codes Set	DTC(s)	8	Unsigned	1	0	0	255	-	-	No of DTCs logged	A	B			
RAM	0202	Number of Trouble Codes Set due to Diagnostic Test	DTC(s)	8	Unsigned	1	0	0	255	-	-	No of DTCs set due to self-test	A	B			
J1-18	7123	Autolamp Analog Input Status	% Full ON	8	Unsigned	100/255	0	0	100	-	-	Used for Twilight sensor [Note] Output range will be \$13 (7%) - \$9A (60%)	-	B			
J4-8	9927	Interior Actual Temperature #2	Degrees C	8	Unsigned	1	-40	-40	215	-	-	Temperature range measured is -20 to +74 Degrees C. Invalid reading is returned as \$FF (215).	-	B			
J2-17	992B	Sunload Sensor A/D Range	Counts	8	Unsigned	1	0	0	255	-	-	Invalid reading shows as \$FF	-	B			
EEPROM	C104	PATS Number of Ignition Key Codes Supported	Counts	8	Unsigned	1	0	0	255	-	-	Reports number of Ignition keys trained	A	B			
EEPROM	E114	Number of Programmed Transmitters	Counts	8	Unsigned	1	0	0	255	-	YES	Number of remote keys trained \$2E used for entering Remote Keypad Training Mode	A	B			

Note: Ignition switch in Accessories or Ignition position

10.5.2 Packeted Encoded Diagnostic PID

PACKETED PIDs - PIDs whose information is a combination of packets of the other types of PIDs																Writeable States	
Conn Pin Num	PID	PIDName	Size (Bytes)	Byte	Bit (s)	'0'	'1'	State	Units	Type	Res	Offset	Min	Max	\$2F	\$2E	
EEPROM	7141	Autolamp Twilight Level Settings	2	1					Level	Signed			-128	127	-	Yes	
				2					Level	Signed			-128	127	-	Yes	
J2-19	A442	Front Wiper Control Mode Select Position & Delay Interval	2	1				01							-	-	
								02							-	-	
								03							-	-	
								04							-	-	
								05							-	-	
								06							-	-	
								07							-	-	
				2				01							-	-	
								02							-	-	
ROM	E200	Software Version Number	3	1	7-4				n/a	Num	1	0	0	15	-	-	
				1	3-0				Months	Num	1	0	0	15	-	-	
				2					Days	Num	1	0	0	255	-	-	
				3					Years	Num	1	1900	1900	2155	-	-	
ROM	E217	ECU Part Number Base	4	1	7-4					BCD					-	-	
				1	3-0					BCD					-	-	
				2						AlphaNum					-	-	
				3						AlphaNum					-	-	
				4	7-4					BCD					-	-	
				4	3-0					BCD					-	-	
ROM	E219	Part Identification Suffix	2	1	7-1					SED					-	-	
				1/2	0(7-5)					BCD					-	-	
				2	4-0					SED					-	-	
ROM	E21A	ECU Part Number ID Prefix	4	1						ASCII					-	-	
				2						ASCII					-	-	
				3						ASCII					-	-	
				4						ASCII					-	-	

10.5.3 State Encoded Diagnostic PID
STATE ENCODED PIDS - PIDs whose information is represented by "1 of n" possible states.

Conn Pin Num	PID	PID Name	State	State Description	Writeable States		Comments/Information	ECU Type		
					\$2F	\$2E				
J2-12, J1-10 J4-7, J4-8	A431	Ignition Switch Position	01	Switch in ACC position	-	-	Transponder is away from ignition. Cannot be determined within required response timeframe	A	B	
			02	Key-Out	-	-		A	B	
J2-12, J1-10			03	Switch in OFF / LOCK position	-	-		A	B	
J2-12, J1-10			04	Switch in RUN position	-	-		A	B	
J2-12, J1-10			05	Switch in START position	-	-		A	B	
J2-12, J4-40			FF	Invalid Data	-	-	Crank position	A	B	
J1-7, J1-8	C125	Anti-Theft Status	01	Immobilised	-	-		A	B	
J1-7, J1-8			02	Armed	-	-		A	B	
J1-7, J1-8			03	Mobilised	-	-		A	B	
EEPROM			10	Plant Mode	-	-		A	B	
EEPROM			11	Transponder Dealer Teach Mode	-	YES	Either PCM or DST plant mode \$2E used to Invoke Transponder Training Mode. Will erase all known transponders and start transponder training process. The \$2E command is only valid if sent within 2 minutes of Ignition switch being turned to the off position.	A	B	
CAN Msg \$726	C801	Intermediate Remote Key Plant Mode	00	Plant mode	-	-	Battery saver time out = 3 minutes	A	B	
			01	Intermediate Remote Key Plant Mode	-	YES	\$2E used to exit plant mode prior to learning remote keys. Battery saver time out = 3 minutes	A	B	
			02	Keypad training performed	-	-	Battery saver time out = 15 minutes	A	B	
INTERNAL	D100	ECU Operating State	81	Standard Diagnostic Mode	-	-	Always in this mode	A	B	
RAM. J4-1,2,3,4	E112	Last Remote Keyless Entry Received Data Byte	00	Button Data Cleared	-	-	Status after BEM power reset until valid data received.	A	B	
			01	Unlock Button - Single stage unlock mode	-	-		A	B	
			02	Decklid Button	-	-		A	B	
			04	Lock Button	-	-		A	B	
			08	Panic Button	-	-		A	B	
			11	Short (simultaneous button press)	-	-		A	B	
			12	Unlock Button - Two stage unlock mode	-	-		A	B	
			FF	Invalid Data	-	-	Not Implemented. Captured by invalid data (\$FF) in PID \$E415.	A	B	
RAM. J4-1,2,3,4	E116	Last Received Transmitter ID Code (TIC) Reference Number (*- \$2F command not supported on low series)	00	Unknown	-	-	433MHz untrained remote key-	A	B	
			01	Key Fob #1	YES *	-	See note below	A	B	
			02	Key Fob #2	YES *	-	See note below	A	B	
			03	Key Fob #3	YES *	-	See note below	A	B	
			04	Key Fob #4	YES *	-	See note below	A	B	
			05	Key Fob #5	YES *	-	See note below	A	B	
			06	Key Fob #6	YES *	-	See note below	A	B	
			07	Key Fob #7	YES *	-	See note below	A	B	
			08	Key Fob #8	YES *	-	See note below	A	B	
			F0	Master Key Fob	YES *	-	See note below	A	B	
			FF	Invalid Data	-	-	There is no known remote key pressed since BEM reset.	A	B	
EEPROM	E11A	Body Type	00	Unknown Body Type	-	-	Configure using block write \$01	A	B	
			01	Sedan	-	-	Configure using block write \$01	A	B	
			02	Wagon	-	-	Configure using block write \$01	A	B	
			03	Utility	-	-	Configure using block write \$01	A	B	
			04	LWB	-	-	Configure using block write \$01	A	B	
			05	4WD	-	-	Configure using block write \$01	A	B	
			06	Coupe	-	-	Configure using block write \$01	A	B	
			07	Van	-	-	Configure using block write \$01	A	B	

10.5.4 Bit Mapped Diagnostic PID

BIT MAPPED PIDS - PIDs whose information is provided in a binary ("1" or "0") bit mapped format.										Writeable States		Comments	ECU Type		
Conn Pin Num	PID	PID Name	Size	Byte	Bit	Bit Description	"0"	"1"	\$2F	\$2E					
J2-1	7121	Autolamp On Output Status	16	1	Bit 7	Autolamp On Driver Output State	Off	On	YES	-	Headlamp output	-	B		
					Bit 6	N/A	x	-	-	-					
					Bit 5	N/A	x	-	-	-					
					Bit 4	N/A	x	-	-	-					
					Bit 3	N/A	x	-	-	-					
					Bit 2	N/A	x	-	-	-					
					Bit 1	N/A	x	-	-	-					
					Bit 0	N/A	x	-	-	-					
				2	Bit 7-0	N/A	x	-	-	-					
	A141	External Access Ajar Input Switch Status	16	1	Bit 7	N/A	x	-	-	-					
					Bit 6	N/A	x	-	-	-					
J1-12					Bit 5	Left Rear Door Ajar	No (False)	Yes (True)	-	-	Passenger door	A	B		
J1-11					Bit 4	Right Rear / Sliding Door Ajar	No (False)	Yes (True)	-	-	Passenger door	A	B		
J1-17					Bit 3	Decklid / Hatch Ajar	No (False)	Yes (True)	-	-	Used for boot/tailgate	A	B		
					Bit 2	N/A	x	-	-	-					
					Bit 1	N/A	x	-	-	-					
					Bit 0	N/A	x	-	-	-					
				2	Bit 7	N/A	x	-	-	-					
					Bit 6	N/A	x	-	-	-					
					Bit 5	N/A	x	-	-	-					
					Bit 4	N/A	x	-	-	-					
					Bit 3	N/A	x	-	-	-					
					Bit 2	N/A	x	-	-	-					
J1-13					Bit 1	Left Front Door Ajar	No (False)	Yes (True)	-	-	Passenger door	A	B		
J1-14					Bit 0	Right Front Door Ajar	No (False)	Yes (True)	-	-	Driver door	A	B		
J2-24	A149	Drivers Doors Unlock Output Status	46	4	Bit 7	Drivers Doors Unlock Driver Output State	Disable(d)	Enable(d)	-	-	Driver's Door Feedback	A	B		
	-	-	-		Bit 6	N/A	x	-	-	-	Use new PID \$A15B				
					Bit 5	N/A	x	-	-	-					
					Bit 4	N/A	x	-	-	-					
					Bit 3	N/A	x	-	-	-					
					Bit 2	N/A	x	-	-	-					
					Bit 1	N/A	x	-	-	-					
					Bit 0	N/A	x	-	-	-					
				2	Bit 7-0	N/A	x	-	-	-					
J5-2	A14B	Primary Lock Output Status	16	1	Bit 7	Driver's Lock Activated	No (False)	Yes (True)	YES	-	\$22 not supported. Lock activation completes within 200ms. Too fast to monitor. Driver door and Passenger doors cannot be locked independently (ie. all doors will be locked).	A	B		
J3-9					Bit 6	Driver's Unlock Activated	No (False)	Yes (True)	YES	-		A	B		
J5-2					Bit 5	Passenger's Lock Activated	No (False)	Yes (True)	YES	-		A	B		
J5-1					Bit 4	Passenger's Unlock Activated	No (False)	Yes (True)	YES	-		A	B		
					Bit 3	N/A	x	-	-	-					
					Bit 2	N/A	x	-	-	-					
					Bit 1	N/A	x	-	-	-					
					Bit 0	N/A	x	-	-	-					
				2	Bit 7-0	N/A	x	-	-	-					
J2-11	A150	Power Window Driver's Output Status	16	1	Bit 7	Window Driver Output State	Disable(d)	Enable(d)	YES	-		A	B		
					Bit 6	N/A	x	-	-	-					
					Bit 5	N/A	x	-	-	-					
					Bit 4	N/A	x	-	-	-					
					Bit 3	N/A	x	-	-	-					
					Bit 2	N/A	x	-	-	-					
					Bit 1	N/A	x	-	-	-					
					Bit 0	N/A	x	-	-	-					
				2	Bit 7-0	N/A	x	-	-	-					
	A15B	All door lock input/output status	16	1	Bit 7-0	N/A	x	-	-	-					
J2-24				2	Bit 7	Drivers door lock status	Unlocked	Locked	-	-		A	B		
					Bit 6	N/A	x	-	-	-					
					Bit 5	N/A	x	-	-	-					
					Bit 4	N/A	x	-	-	-					
					Bit 3	N/A	x	-	-	-					
					Bit 2	N/A	x	-	-	-					
					Bit 1	N/A	x	-	-	-					
					Bit 0	N/A	x	-	-	-					
J3-7	A200	Decklid Output Status	16	1	Bit 7	Decklid Driver Output State	Off	On	YES	-	Boot solenoid driver	A	B		
					Bit 6	N/A	x	-	-	-					
					Bit 5	N/A	x	-	-	-					
					Bit 4	N/A	x	-	-	-					
					Bit 3	N/A	x	-	-	-					

J1-9						Bt 2 Bt 1 Bt 0	N/A N/A Trunk Release Input Switch	x x Inactive	--- --- Active	- - -	- - -	Read switch status	A B
				2	Bt 7-0	N/A		x	---				
J1-20	A445	Windshield Wiper / Washer Input Switch Status	16	1	Bt 7 Bt 6 Bt 5	Washer Fluid Level N/A N/A	Low x x	OK --- ---	- - -	- - -	Filtered(20s) Status	A B	
J3-2					Bt 4 Bt 3	Wiper Motor Status N/A	Inactive x	Active ---	- -	- -	Read status	A B	
J3-1					Bt 2 Bt 1 Bt 0	N/A N/A Wiper in Park Position	x x No (False)	--- --- Yes (True)	- - -	- - -	Read status	A B	
				2	Bt 7-0	N/A		x	---				
J3-2	A452	Wiper Motor Run & Two Speed Output Status	16	1	Bt 7 Bt 6 Bt 5 Bt 4 Bt 3 Bt 2 Bt 1 Bt 0	Wiper Run Driver Output State N/A N/A N/A N/A N/A N/A N/A N/A	Off x x x x x x x	On --- --- --- --- --- --- ---	YES - - - - - - -	- - - - - - - -	Control wiper [Note]	A B	
					Bt 7-0	N/A		x	---				
J3-10	A453	Washer Pump Output Status	16	1	Bt 7 Bt 6 Bt 5 Bt 4 Bt 3 Bt 2 Bt 1 Bt 0	Washer Pump Driver Output State N/A N/A N/A N/A N/A N/A Washer Pump Switch	Off x x x x x x Inactive	On --- --- --- --- --- --- Active	YES - - - - - - -	- - - - - - -	Control washer (3 wipes w/i follow wash). \$22 not supported. Washer pump switch is wired (connected) to the washer pump relay output [note]	- B	
				2	Bt 7-0	N/A		x	---				
J3-12	A455	Illuminated Entry & Battery Saver Output Status	16	1	Bt 7 Bt 6 Bt 5 Bt 4 Bt 3 Bt 2 Bt 1 Bt 0	Illuminated Entry Driver Output State N/A N/A N/A Battery Saver Driver Output State N/A N/A N/A Heated Backlite Switch	Off x x x x x x Inactive	On --- --- --- --- --- --- Active	YES - - - - - - -	- - - - - - -	Control / Read dome lamp output status, Control with Battery Saver Circuit.	A B	
J3-3					Bt 7 Bt 6 Bt 5 Bt 4 Bt 3 Bt 2 Bt 1 Bt 0	Battery Saver Driver Output State N/A N/A N/A N/A N/A N/A N/A Heated Backlite Switch	Off x x x x x x Inactive	On --- --- --- --- --- --- Active	YES - - - - - - -	- - - - - - -	Control / Read Battery Saver output status	A B	
				2	Bt 7-0	N/A		x	---				
J2-10	A457	Heated Backlight Output Status	16	1	Bt 7 Bt 6 Bt 5 Bt 4 Bt 3 Bt 2 Bt 1 Bt 0	Heated Backlite Driver Output State N/A N/A N/A N/A N/A N/A N/A Heated Backlite Switch	Off x x x x x x Inactive	On --- --- --- --- --- --- Active	YES - - - - - - -	- - - - - - -	Control demist.	A B	
CAN Msg #307					Bt 7 Bt 6 Bt 5 Bt 4 Bt 3 Bt 2 Bt 1 Bt 0	Heated Backlite Switch N/A N/A ACM Instrument Cluster (IC) N/A N/A N/A Powertrain Control Module	Inactive x x x No (False) No (False) x x No (False)	Active ---	- - - - - - - -	- - - - - - -	Read switch status via CAN	A B	
				2	Bt 7-0	N/A		x	---				
B802	Fault PID (CM): Missing CAN message source id	32	1	Bt 7 Bt 6 Bt 5 Bt 4 Bt 3 Bt 2 Bt 1 Bt 0	N/A N/A N/A ACM Instrument Cluster (IC) N/A N/A N/A Powertrain Control Module	x x x x No (False) No (False) x x No (False)	---			- - - - - - -	- - - -	CDU Msg #307 Cluster Msg #437	A B A B
J2-13,14,15,16					Bt 7 Bt 6 Bt 5 Bt 4 Bt 3 Bt 2 Bt 1 Bt 0	N/A N/A N/A ACM Instrument Cluster (IC) N/A N/A N/A Powertrain Control Module	No (False) x x x No (False) No (False) x x No (False)	Yes (True) ---	- - - - - - -	- - - -	PCM Msg #207, #427	A B	
J2-13,14,15,16				2-4	Bt 7-0	N/A		x	---				
J2-9	C110	Horn Control Output Status	16	1	Bt 7 Bt 6 Bt 5 Bt 4 Bt 3 Bt 2 Bt 1 Bt 0	Horn Control Driver Output State N/A N/A N/A N/A N/A N/A N/A Horn Control Switch	Off x x x x x x Inactive	On --- --- --- --- --- --- Active	YES - - - - - - -	- - - - - - -		A B	

Note: For writable state \$2F, Ignition switch in Accessories or Ignition position

	C124	Passive Anti-Theft System Status	16	1	Bit 7	N/A	-	-	-	-	
					Bit 6	N/A	-	-	-	-	
					Bit 5	N/A	-	-	-	-	
					Bit 4	N/A	-	-	-	-	
J2-13,14,15,16					Bit 3	PCM Verify OK	No (False)	Yes (True)	-	-	Acknowledgement received
J2-13,14,15,16					Bit 2	PCM ID Status Stored	No (False)	Yes (True)	-	-	Plant mode
J1-3,4					Bit 1	Anti-Scan Function	Off	On	-	-	
J2-13,14,15,16					Bit 0	PCM Request Received	No (False)	Yes (True)	-	-	Challenge Received
				2	Bit 7-0	N/A	x	---	-	-	
	C900	BEM-PCM training step	16	1	Bit 7	N/A	x	---	-	-	
CAN Msg \$6F6					Bit 6	PCM verified response to be successful	No (False)	Yes (True)	-	-	
CAN Msg \$6F8					Bit 5	BEM sent first response to PCM	No (False)	Yes (True)	-	-	
CAN Msg \$6F6					Bit 4	BEM received valid PCM challenge	No (False)	Yes (True)	-	-	
CAN Msg \$6F6					Bit 3	BEM received PCM challenge	No (False)	Yes (True)	-	-	
CAN Msg \$6F6					Bit 2	BEM received \$6F6 from PCM	No (False)	Yes (True)	-	-	
Internal					Bit 1	BEM Exited PCM/BEM plant mode	No (False)	Yes (True)	-	-	
Internal					Bit 0	BEM learnt/stored seed	No (False)	Yes (True)	-	-	
CAN Msg \$6F6				2	Bit 7	PCM acknowledged BEM learnt PCM seed	No (False)	Yes (True)	-	-	
CAN Msg \$6F8					Bit 6	BEM acknowledgement sent to PCM	No (False)	Yes (True)	-	-	
Internal					Bit 5	BEM received valid PCM seed	No (False)	Yes (True)	-	-	
CAN Msg \$6F6					Bit 4	BEM received PCM seed	No (False)	Yes (True)	-	-	
CAN Msg \$6F6					Bit 3	BEM received \$6F6 from PCM	No (False)	Yes (True)	-	-	
TILA comms					Bit 2	TILA/BEM plant mode exited	No (False)	Yes (True)	-	-	
TILA comms					Bit 1	More than 1 transponder was trained	No (False)	Yes (True)	-	-	
TILA comms					Bit 0	One transponder was trained successfully	No (False)	Yes (True)	-	-	
	EEPROM	E11B	Personalization Settings	32	1	Bit 7	Taxi Mode	Disable	Enable	-	Configure using block write \$00
					Bit 6	Panic Alarm Inhibit	Disable	Enable	-	-	Configure using block write \$00
					Bit 5	Door Trigger Disable Mode	Disable	Enable	-	-	Configure using block write \$00
					Bit 4	Master Door Trigger Disable Mode	Disable	Enable	-	-	Configure using block write \$00
					Bit 3	Hazard Warning Inhibit	Disable	Enable	-	-	Configure using block write \$00
					Bit 2	Horn Warning Inhibit	Disable	Enable	-	-	Configure using block write \$00
					Bit 1	Battery Saver Inhibit	Disable	Enable	-	-	Configure using block write \$00
					Bit 0	Courtesy Lamp Inhibit	Disable	Enable	-	-	Configure using block write \$00
				2-4	Bit 7-0	N/A	x	---	-	-	

Note:

 CAN connector pins are J2-13,14,15,16.
 TILA connector pins are J1-3,4, J1-7,8.

10.5.5 ASCII and BCD Diagnostic PID

ASCII PIDs - PIDs whose information is provided as ASCII data

Conn Pin Num	PID	PID Name	Size (Bytes)	Byte	Bit (s)	"0"	"1"	State	Units	Type	Res	Offset	Min	Max	Writeable States		Data/Bit/State Description	ECU Type	
															\$2F	\$2E		A	B
EEPROM	E300	Vehicle ID (VIN) #1	4	1						ASCII					-	-	VIN character 1. Written using block \$02	A	B
				2						ASCII					-	-	VIN character 2. Written using block \$02	A	B
				3						ASCII					-	-	VIN character 3. Written using block \$02	A	B
				4						ASCII					-	-	VIN character 4. Written using block \$02	A	B
EEPROM	E301	Vehicle ID (VIN) #2	4	1						ASCII					-	-	VIN character 5. Written using block \$03	A	B
				2						ASCII					-	-	VIN character 6. Written using block \$03	A	B
				3						ASCII					-	-	VIN character 7. Written using block \$03	A	B
				4						ASCII					-	-	VIN character 8. Written using block \$03	A	B

BCD PIDs - PIDs whose information is provided as Binary-Coded-Decimal data

Conn Pin Num	PID	PID Name	Size (Bytes)	Byte	Bit (s)	"0"	"1"	State	Units	Type	Res	Offset	Min	Max	Writeable States		Data/Bit/State Description	ECU Type	
															\$2F	\$2E		A	B
EEPROM	E231	Module Serial Number 1 (BCD)	4	1	7-4					BCD					-	-	Digit 1	A	B
				1	3-0					BCD					-	-	Digit 2	A	B
				2	7-4					BCD					-	-	Digit 3	A	B
				2	3-0					BCD					-	-	Digit 4	A	B
				3	7-4					BCD					-	-	Digit 5	A	B
				3	3-0					BCD					-	-	Digit 6	A	B
				4	7-4					BCD					-	-	Digit 7	A	B
				4	3-0					BCD					-	-	Digit 8	A	B
EEPROM	E232	Module Serial Number 2 (BCD)	4							BCD					-	-	Return \$FFFFFFFF	A	B
EEPROM	E233	Module Serial Number 3 (BCD)	4							BCD					-	-	Return \$FFFFFFFF	A	B
EEPROM	E234	Module Serial Number 4 (BCD)	4							BCD					-	-	Return \$FFFFFFFF	A	B
EEPROM	E235	Module Serial Number 5 (BCD)	4							BCD					-	-	Return \$FFFFFFFF	A	B
EEPROM	E236	Module Serial Number 6 (BCD)	4							BCD					-	-	Return \$FFFFFFFF	A	B



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ENGINEERING SPECIFICATION

FOA Barra Body Electronics Module
High and Low Series

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10.6 Diagnostic Trouble Codes (DTC)

DTCs		for: IDENTIFYING & REPORTING SUBSYSTEM FAULTS			for additional information, refer to: Section 9.0 - Diagnostic Trouble Code			
Form Version 4.0		Include all mandatory DTCs (including network faults)			(DTC) Requirements of the Global Diagnostic Specification (Part 1)			
NOTE: DTC information can be copied directly from the Master Ref. database and pasted into this specification.								
WHEN SET Key: "C": Continuous Operation								
"D1": set due to self test request \$02.								
"D2": set due to self test request \$11.								
Conn Pin Num	DTC	Display	Description	How SET	Fault Trigger Conditions	Comments/Information (SYSTEM)	ECU Type	
CAN Msg #427	5943	C1943	Airbag Deployment Indication Input Fault	C	D1, D2	Airbag Deployed CAN bit received. Status read during self tests.	Airbag deployed	A B
EEPROM	9213	B1213	Anti-Theft Number of Programmed Keys is Below Minimum	C	D1, D2	Number of programmed keys is less than 2.	PATS	A B
J4-8,9	9250	B1250	Air Temperature Internal Sensor Circuit Failure	C	D1, D2	Out of range. Will be set if input is between 0.32V and 0.47V	Cabin Temperature Sensor [Note 2]	- B
J4-8,9	9251	B1251	Air Temperature Internal Sensor Circuit Open	C	D1, D2	Input floats > 4.56V	Cabin Temperature Sensor [Note 2]	- B
J4-8,9	9253	B1253	Air Temperature Internal Sensor Circuit Short To Ground	C	D1, D2	Input < 0.32V, External Fault	Cabin Temperature Sensor [Note 2]	- B
-	9258	B1258	Solar Radiation Sensor Circuit Failure	C	D1-D2	Removed because there is no difference to open circuit condition	Driver's Sunload Sensor	- B
J2-6, J2-17	9259	B1259	Solar Radiation Sensor Circuit Open	C	D1, D2	Circuit open or short to battery, Input greater than 4.76V, External Fault	Driver's Sunload Sensor [Note 2]	- B
J2-6, J2-17	9261	B1261	Solar Radiation Sensor Circuit Short To Ground	C	D1, D2	Input held at less than 0.17V, External Fault	Driver's Sunload Sensor [Note 2]	- B
J5-2, J2-24	9300	B1300	Power Door Lock Circuit Failure	C	D1, D2	Door Lock Motor Error / Driver's door lock feedback did not change after lock action, External Fault	Door Locks	A B
INTERNAL	9342	B1342	ECU Is Faulted	C	D1, D2	EEPROM / MEMIC are continuous only, but if set the DTC will be reported as part of self test. ROM / RAM DTC can be set after BEM power reset (continuous DTC) or during self test - Internal Fault	ECU Failure - Replace Module.	A B
J2-10	9348	B1348	Heated Backlite Relay Circuit Open	C	D1, D2	O/C or S/C to Ground, External Fault	Heated Backlite [Note 1]	A B
J2-10	9349	B1349	Heated Backlite Relay Short To Battery	C	D1, D2	S/C to Battery, External Fault	Heated Backlite [Note 1]	A B
J3-1,2	9446	B1446	Wiper Park Sense Circuit Failure	C	D1, D2	Feedback does not change / park circuit failure, External Fault	Wiper [Note 1]	A B
J3-10	9458	B1458	Wiper Washer Pump Motor Relay Circuit Failure	C	D1, D2	Front wash motor error, External Fault	Wiper / Washer [Note 1]	- B
J2-11	9548	B1548	Power Window Master Circuit Open	C	D1, D2	Power Window Relay driver O/C or S/C to Ground, External Fault	Power Windows	A B
J2-11	9549	B1549	Power Window Master Circuit Short To Battery	C	D1, D2	Power Window Relay driver S/C to Battery, External Fault	Power Windows	A B
TILA, J1-7	9600	B1600	PATS Ignition Key Transponder Signal is Not Received	C	D1, D2	TILA RF receiver error	PATS	A B
TILA	9601	B1601	PATS Received Incorrect Key-Code From Ignition Key Transponder	C	D1, D2	Key ID unknown or incorrect challenge	PATS	A B
TILA	9602	B1602	PATS Received Invalid Format Of Key-Code From Ignition Key Transponder	C	D1, D2	Faulty or fixed code transponder	PATS	A B
-	9676	B1676	Battery-Pack Voltage Out-of-Range	-	-	Hardware overvoltage detection is removed, and BEM not monitoring or reading value of battery voltage.	-	- -
J1-5,6	9693	B1693	Autolamp On Circuit Failure	C	D1, D2	Autoheadlamps Driver Circuit Failure, External Fault	Twilight sensor Headlamp switch must be in Auto position.	- B
J1-18	9790	B1790	Autolamp Sensor Input Circuit Failure	C	D1, D2	Twilight sensor input circuit failure	Twilight sensor [Note 1]	- B
J1-18	9791	B1791	Autolamp Sensor Input Circuit Open	C	D1, D2	Twilight sensor input circuit open, External Fault	Twilight sensor	- B
J2-24, J3-9	9882	B1882	Driver's Door Unlock Relay Circuit Failure	C	D1, D2	Driver's door unlock is not successful / Driver's door lock feedback did not change after unlock action	Door Locks	- B
J2-22	A072	B2072	Seat Memory Module Data Line Short to Battery	C	D1, D2	S/C to Battery, External Fault	Memory Seat Module when BEM is sending seat memory data	- B
J2-22	A073	B2073	Seat Memory Module Data Line Open or Short to Ground	C	D1, D2	O/C or S/C to Ground, External Fault	Memory Seat Module when BEM is sending seat memory data	- B
J2-23	A074	B2074	Seat Memory Module Enable Line Short to Battery	C	D1, D2	S/C to Battery, External Fault	Memory Seat Module when BEM is sending seat memory data	- B
J2-23	A075	B2075	Seat Memory Module Enable Line Open or Short to Ground	C	D1, D2	O/C or S/C to Ground, External Fault	Memory Seat Module when BEM is sending seat memory data	- B
TILA	A103	B2103	Antenna Not Connected	C	D1, D2	TILA antenna not connected, External Fault	PATS	A B
EEPROM	A141	B2141	NVM Configuration Failure	C	D1, D2	No PCM ID Stored	PATS	A B
CAN Msg #6F6	A431	B2431	Transponder Programming Failed	C	-	Key programming failed	PATS	A B
EEPROM	A477	B2477	Module Configuration Failure	C	D1, D2	Calibration factor is not configured	Programming	- B
J2-9	A511	B2511	Horn Output Relay Circuit Short to Battery	C	D1, D2	S/C to Battery, External Fault	Horn	A B
CAN Msg #6F6	A682	B2682	EEC (PCM) Communication Fault	C	*D1, *D2	Immobilisation Message Invalid	PATS	A B
J2-9	A640	B2640	Horn Output Relay Circuit Open	C	D1, D2	O/C or S/C to Ground, External Fault	Horn	A B
J2-19	A891	B2891	Front Wiper Switch Potentiometer Circuit Failure	C	D1, D2	Potentiometer is missing, External Fault (O/C or S/C)	Wiper [Note 1]	A B
J2-19	A892	B2892	Front Wiper Switch Potentiometer Out-of-Range	C	D1-D2	Potentiometer reading is out-of-range, External Fault	Wiper	- B
J3-7, J1-17	A894	B2894	Deadlock Release Circuit Failure	C	D1, D2	Boot is not ajar after boot release, External Fault	Boot release solenoid error	A B
-	A928	B2928	RF Receiver DTC Rx not connected	-	-	Removed due to unreliable detection - Cannot determine difference between no key fob present and an open circuit.	-	- -
J1-8	A929	B2929	Transmit Data Line Short to Battery	C	D1, D2	S/C to Battery, External Fault	PATS when BEM is challenging the transponder	A B
J1-8	A930	B2930	Transmit Data Line Open Circuit or Short to Ground	C	D1, D2	O/C or S/C to Ground, External Fault	PATS when BEM is challenging the transponder	A B
CAN BUS	D900	U1900	CAN Communication Bus Fault - Receive Error	C	D1, D2	CAN Network failure (Missing Messages)	Used with PID \$B002	A B
CAN Msg #207	E197	U2197	Invalid Vehicle Speed data	C	D1, D2	CAN bytes for vehicle speed is read as 'FF FF'	CAN Message from PCM #207 [Note 2]	A B

Note:

CAN connector pins are J2-13,14,15,16.

TILA connector pins are J1-3,4, J1-7,8.

* Self Test will detect the error only if EEC sends an Immobilisation Message.

Note 1: Ignition switch in Accessories or Ignition position

Note 2: Ignition switch in Ignition position

10.7 Module Configuration via Request Read/Write Memory Block

Method 2: Module Configuration via REQUEST READ MEMORY BLOCK [\$21] / REQUEST WRITE MEMORY BLOCK [\$3B]:

NUMERIC WRITE MEMORY BLOCK

Block Num	Command Description	Unit	Size	Type	Offset	Resolution	MIN	MAX	Default	Comments/Information	ECU Type
\$0E	Cabin Temperature Sensor Calibration Factor	Bit	3-0	Signed	N/A	1	-6	6	\$F	Offset Factor used to calibrate CTS readings	- B
		Bit	7-4	-	-	-	-	-	\$F	Set to \$F as default	- B

STATE ENCODED WRITE MEMORY BLOCK

Block Num	Command Description	State	State Description	Default	Disabled Diagnostic Elements Note: List elements not supported when feature disabled					Comments/Information	ECU Type
					CMD(s)	PID(s)	DTC(s)				
\$01	Body Type	Byte	1	7-0	-	-	-	-	-	Refer SED PID \$E11A for details	A B

BIT MAPPED WRITE MEMORY BLOCK

Block Num	Command Description	Size	Byte	Bit	Bit Description	"0"	"1"	Default		Comments/Information	ECU Type
\$00	Police/Taxi Mode Control Byte	4	1	7-0	-	-	-	-	-	Refer BMP PID \$E11B for details	A B

ASCII WRITE MEMORY BLOCK

Block Num	PID	PID Name	Size (Bytes)	Byte	Bit (s)	"0"	"1"	State	Units	Data/Bit/State Description	ECU Type
\$02	VIN Number (First Four Characters)		4	1-4	7-0	-	-	-	-	Refer ASCII PID \$E300 for details	A B
\$03	VIN Number (Last Four Characters)		4	1-4	7-0	-	-	-	-	Refer ASCII PID \$E301 for details	A B

PACKETED WRITE MEMORY BLOCK

Block Num	PID	PID Name	Size (Bytes)	Byte	Bit (s)	"0"	"1"	State	Units	Data/Bit/State Description	ECU Type
\$04	Remote Key Settings - Block 1		5	1-5	7-0	-	-	-	-	Encrypted keypad data. Allows trained remote keys to be transferred to new BEM without going through training procedure again. (50 bytes of data needed for 8 remote keys)	A B
\$05	Remote Key Settings - Block 2		5	1-5	7-0	-	-	-	-		A B
\$06	Remote Key Settings - Block 3		5	1-5	7-0	-	-	-	-		A B
\$07	Remote Key Settings - Block 4		5	1-5	7-0	-	-	-	-		A B
\$08	Remote Key Settings - Block 5		5	1-5	7-0	-	-	-	-		A B
\$09	Remote Key Settings - Block 6		5	1-5	7-0	-	-	-	-		A B
\$0A	Remote Key Settings - Block 7		5	1-5	7-0	-	-	-	-		A B
\$0B	Remote Key Settings - Block 8		5	1-5	7-0	-	-	-	-		A B
\$0C	Remote Key Settings - Block 9		5	1-5	7-0	-	-	-	-		A B
\$0D	Remote Key Settings - Block 10		5	1-5	7-0	-	-	-	-		A B
\$0F	Twilight Sensor Setting		2	1	7-0	-	-	-	-	ON Threshold for autoheadlamp. Refer PID-PKT \$7141	- B
			2	2	7-0	-	-	-	-	OFF Thresholds for autoheadlamp. Refer PID-PKT \$7141	- B

Note:

\$10 - \$FF is out of range for high series. Return code \$31.

\$0E - \$FF is out of range for low series. Return code \$31.

11. APPENDIX B – CAN Message List

11.1 Message Overview

#	Message Name	Source	Message Identifier	Data								RX's					TX Model			Data Type									
				1	2	3	4	5	6	7	8	PCM	ACM	ABS/TCS	BEM	HIM	IC	Periodic	Event	Hardware Wakeup	PRN	Type							
				16 Bit Word MSByteLSByte Bit 63Bit 0																									
1	Torque Data and Engine Flags	PCM	200	Actual Torque		Minimum Torque		Maximum Torque		Engine Flags		Reserved				X				50									
2	Engine RPM with high resolution Rate of Change	PCM	207	Engine RPM		RPM Rate of change		Vehicle Speed		Throttle Pos		Throttle Pos ROC			X	X	X	X	X	50									
3	Desired Torque, Brake Configuration and Telltales	TCS	210	Desired Torque		Brake Sys Config		Telltales		Reserved				X				X	20										
4	Transmission Gear, Torque converter effects, trans losses & trans flags	PCM	230	Gear Position		Gear Ratio		Torque converter torque multiplication		Transmission losses/mode		Trans Flags/ Mode			X	X			X	64									
5	Audio climate and auxilliary switches	ACM	307	Climate Control				Aux Switches		Trip Comp Sw		Man Temperature		Man_Fan		CDU type													
6	HVAC Climate Control Status	HIM	353	Display Data		ACC Display Data		Pass Temp		Drv Temp		Amb Temp		Evap Temp		Blower Volt		Flags											
7	BEM Status	BEM	403	Switch Status		Headlamp/ Priority Key		Cabin Temp		Drv/Sunload		Pas/Sunload		Pr. Key/ Reserved		Reserved			X										
8	Powertrain Driver Information	PCM	427	ECT		ODO Count		Fuel Pulse		Battery Volt		DFLPG Tank Lev		Telltales		Reserved			X		X	X							
9	Fuel and Warnings	IC	437	Fuel Level		Inst Fuel		Warnings		Battery Voltage		Illumination - Display		Illumination - Switch		Reserved			X										
10	Wheel Speed	TCS	480	Front Left Wheel Speed			Front Right Wheel Speed			Rear Left Wheel Speed			Rear Right Wheel Speed			Travel time													
11	Trip Computer Data	IC	500	Ave Speed		Ave Economy		Reserved		Reserved		Reserved							X										
12	Trip Computer Data	IC	553	Range			Instantaneous Economy			Odometer			Reserved						X	X									
13	Engine Configuration	PCM	623	Engine Displ.		Cylinders & Valves		Fuel Type & Deliv/y		Peak Torque		Aux Veh ID		Reserved						X									
14	Transmission Configuration	PCM	640	Trans/Axle Type		Axle Ratio Front			Fractional Portion of Axle Ratio			Base Torque Split		Reserved						X									
15	PCM immobilisation message	PCM	6F6	Reserved								Reserved									X			X		250	No		
16	BEM immobilisation transmission	BEM	6F8	Reserved								Reserved									X						Yes		

N.B. This message overview is indicative only, for details refer to individual message structures on the following pages.

ACM - Audio Control Module
PCM - Powertrain Control Module
BEM - Body Electronics Module
ABS/TCS - Antilock Braking, with Traction Control
HIM - HIM Integrated Module
IC - Instrument Cluster

*1 Events result in an immediate message transmission and the next message is transmitted after 500ms resuming the 500 ms periodic transmissions, unless a new event occurs.
(Event messages sent with Ignition off (CAN wake-up active) will be sent once only, or until transmitting module receives confirmation of successful transmission)
Refer to Section 6 document for details

The following are deviations from Ford Multiplex Strategy and apply to Barra:
ID \$210 Desired Torque - 20ms period is a deviation from the generic 50ms
ID \$480 Wheel Speeds - 20ms period is a deviation from the generic 10ms

Multi-byte parameters are stored in "Motorola Format". For example, the 16 bit Engine RPM parameter is stored high byte (bits 15-8) in byte 1 and low byte (bits 7-0) in byte 2 and are located in bits 63 to 47.

11.2 Message \$207 Powertrain Control Module

PRN #	Length	Transmitter	Period	Message Description
207	64	PCM	50ms	Engine RPM with high resolution Rate of Change

Signal Name	Bits Used		Full Range						Multipliy g Factor	Offset	Units or Meaning	Data Type	Utilized Range		Receiving ECU								Valid During			
			Hexadecimal		Invalid Data	Converted		Decimal					ACM	BSM	Hw	K	M	PCM	TCS	Crank	Ign	Acc	S			
	Min	Max	Min	Max		Min	Max																			
Vehicle speed (Multiplying factor = 1/128)	31	16	\$0000	\$FFFE	\$FFFF	0	511.98	0.0078125	0		kph	Unsigned Numeric	0	511.98	R	R	R	R				Y	Y	N/A	N/A	

X ms = transmit
 Y = Sent
 N = Not Sent
 I = Invalid Data Sen
 N/A = Not Applicable

11.3 Message \$307 Audio Control Module

PRN #	Length	Transmitter	Period	Message Description
307	64	ACM	600ms	Audio climate and auxiliary switches

Signal Name	Bits Used		Full Range						Multipliy g Factor	Offset	Units or Meaning	Data Type	Utilised Range		Receiving ECU								Valid During				
			Hexadecimal		Invalid Data	Converted							Decimal		ACM	DEM	HIM	IS	PCM	TCS	Crank	Ign	Acc	Off	Wake Up		
	Min	Max	Min	Max		Min	Max						Min	Max													
Rear demist switch status	61	61	0	1	n/a	Off	On	n/a	n/a	n/a	Bit-mapped				R												
Illumination / Battery Saver re	39	39	0	1	n/a	Off	On	n/a	n/a	n/a	Bit-mapped				R		R										
Door lock switch status	38	38	0	1	n/a	Off	On	n/a	n/a	n/a	Bit-mapped				R												
Dome lamp switch status	37	37	0	1	n/a	Off	On	n/a	n/a	n/a	Bit-mapped				R												

X ms = transmit
 Y = Sent
 N = Not Sent
 I = Invalid Data
 N/A = Not Applicable

11.4 Message \$403 BEM

PRN #	Length	Transmitter	Period	Message Description
403	64	BEM	600ms	BEM Status

Signal Name	Bits Used		Full Range					Multipl g Factor	Offset	Units or Meaning	Data Type	Utilised Range		Receiving ECU								Valid Du			Comments
	MS Bit	LS Bit	Min	Max	Invalid Data	Converted Min	Max					Min	Max	ACK	BEM	HIM	IC	PCM	TCS	Crank	Ign	Acc			
Right front door ajar	63	63	0	1	n/a	No	Yes				Bit-mapped					R	R			Y	Y	Y			Yes = turn on jewel/dome lamp
Left front door ajar	62	62	0	1	n/a	No	Yes				Bit-mapped					R	R			Y	Y	Y			Yes = turn on jewel/dome lamp
Right rear door ajar	61	61	0	1	n/a	No	Yes				Bit-mapped					R	R			Y	Y	Y			Yes = turn on jewel/dome lamp
Left rear door ajar	60	60	0	1	n/a	No	Yes				Bit-mapped					R	R			Y	Y	Y			Yes = turn on jewel/dome lamp
Boot/tailgate open	59	59	0	1	n/a	No	Yes				Bit-mapped					R	R			Y	Y	Y			Yes = turn on jewel/dome lamp
Rear demist segment	58	58	0	1	n/a	No	Yes				Bit-mapped			R						Y	Y	Y			
Accessories switch	57	57	0	1	n/a	Off	On				Bit-mapped					R				Y	Y	Y			
Low washer switch	56	56	0	1	n/a	No	Yes				Bit-mapped					R				Y	Y	Y			'0' Sent on Low Series
Reserved	55	55																							Always 0
Headlamp fail warning	54	54	0	1	n/a	No	Yes				Bit-mapped					R				Y	Y	Y			
Hazard on request	53	53	0	1	n/a	No	Yes				Bit-mapped					R				Y	Y	Y			
Priority key 4	52	52	0	1	n/a	No	Yes				Bit-mapped					R				Y	Y	Y			
Priority key 3	51	51	0	1	n/a	No	Yes				Bit-mapped					R				Y	Y	Y			
Priority key 2	50	50	0	1	n/a	No	Yes				Bit-mapped					R				Y	Y	Y			
Priority key 1	49	49	0	1	n/a	No	Yes				Bit-mapped					R				Y	Y	Y			
Smart Shield Jewel on Request	48	48	0	1	n/a	No	Yes				Bit-mapped					R				Y	Y	Y			
Cabin temperature	47	40	\$00	\$FA	\$FF	-40	85	0.5	-40	°C	Numeric	-40	85			R				N	Y	N			'\$FF' Sent on Low Series
Sunload sensor Driver	39	32	\$00	\$FE	\$FF	0	254	1	0	A/D reading	Numeric	0	254			R				N	Y	N			'\$FF' Sent on Low Series
Sunload sensor Passenger	31	24	\$00	\$FE	\$FF	0	254	1	0	A/D reading	Numeric	0	254			R				N	Y	N			'\$FF' Sent on Low Series
Priority key 8	23	23	0	1	n/a	No	Yes				Bit-mapped					R				Y	Y	Y			
Priority key 7	22	22	0	1	n/a	No	Yes				Bit-mapped					R				Y	Y	Y			
Priority key 6	21	21	0	1	n/a	No	Yes				Bit-mapped					R				Y	Y	Y			
Priority key 5	20	20	0	1	n/a	No	Yes				Bit-mapped					R				Y	Y	Y			
Reserved for wipers on	19	19	0	1	n/a	No	Yes				Bit-mapped									Y	Y	Y			Always 0
Buzzer Request	18	18	0	1	n/a	No	Yes				Bit-mapped									Y	Y	Y			
Reserved	17	0																							Always 0

X ms = transmits for up to Xms
 Y = Sent
 N = Not Sent
 I = Invalid Data Sent
 N/A = Not Applicable

